

OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP - 10: PAGE REPLACEMENT ALGORITHM - FIFO

```
user123@localhost: ~/page
[user123@localhost ~]$ ls
abc.sh  awk  Desktop  Downloads  ifelse.sh  memory  page  Pictures  Public  REC  sample.sh  scheduling  SFA  system_calls  Videos
ab.sh  bankers  Documents  Folder  loops.sh  Music  pc  posneg.sh  qns.txt  sample.c  sample.txt  script  sh  Templates  while.sh
[user123@localhost ~]$ cd page
[user123@localhost page]$ ls
a.out  page_fifo.c
[user123@localhost page]$ vi page_fifo.c
[user123@localhost page]$
```

```
user123@localhost:~/page — /usr/bin/vim page_fifo.c
#include <stdio.h>

int main(){
    int pages[50], frames[10];
    int n, f, i, j, k, found, pageFaults=0;

    //Input
    printf("Enter number of pages: ");
    scanf("%d", &n);

    printf("Enter page reference string: ");
    for(int i=0; i<n; i++){
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: ");
    scanf("%d", &f);

    //Initialize frames
    for(i=0; i<f; i++){
        frames[i] = -1;
    }

    int pos = 0; // FIFO pointer

    printf("\nPage\tFrames\n");

    //FIFO Logic
    for(int i=0; i<n; i++){
        found = 0;

        // check if page already exists
        for(j=0; j<f; j++){
            if(frames[j] == pages[i]){
                found = 1;
                break;
            }
        }

        //if not found -> page fault
        if(found==0){
            frames[pos] = pages[i];
            pos = (pos+1)%f; // circular update
            pageFaults++;
        }

        //print frames
        printf("%d\t", pages[i]);
        for(k=0; k<f; k++){
            if(frames[k] != -1)
                printf("%d ", frames[k]);
            else
                printf("- ");
        }

        printf("\n");
    }
    printf("\nTotal Page Faults = %d\n", pageFaults);

    return 0;
}
```

```
user123@localhost:~/page
[user123@localhost ~]$ ls
abc.sh  awk  Desktop  Downloads  ifelse.sh  memory  page  Pictures  Public  REC  sample.sh  scheduling  SFA  system_calls  Videos
ab.sh  bankers  Documents  Folder  loops.sh  Music  pc  posneg.sh  qns.txt  sample.c  sample.txt  script  sh  Templates  while.sh
[user123@localhost ~]$ cd page
[user123@localhost page]$ ls
a.out  page_fifo.c
[user123@localhost page]$ vi page_fifo.c
[user123@localhost page]$ gcc page_fifo.c
[user123@localhost page]$ ./a.out
Enter number of pages: 5
Enter page reference string: 1 2 3 4 1
Enter number of frames: 3

Page    Frames
1       1 --
2       1 2 -
3       1 2 3
4       4 2 3
1       4 1 3

Total Page Faults = 5
[user123@localhost page]$
```

OPERATING SYSTEM CONCEPTS LAB EXPERIMENTS

EXP - 10: PAGE REPLACEMENT ALGORITHM - LRU

```
user123@localhost:~/page
[user123@localhost ~]$ ls
abc.sh  awk  Desktop  Downloads  ifelse.sh  memory  page  Pictures  Public  REC  sample.sh  scheduling  SFA  system_calls  Videos
ab.sh  bankers  Documents  Folder  loops.sh  Music  pc  posneg.sh  qns.txt  sample.c  sample.txt  script  sh  Templates  while.sh
[user123@localhost ~]$ cd page
[user123@localhost page]$ ls
a.out  page_fifo.c
[user123@localhost page]$ vi page_lru.c
[user123@localhost page]$ gcc page_lru.c
[user123@localhost page]$ ./a.out
Enter number of pages in reference string:
17
Enter the reference string:
7 0 1 2 0 3 0 4 2 3 0 3 2 1 2 0 1
Enter number of frames:
3

Total Page Faults: 11
Total Page Hits: 6
[user123@localhost page]$
```

```
user123@localhost:~/page — /usr/bin/vim page_lru.c
#include <stdio.h>

int main(){
    int n, frames;

    printf("Enter number of pages in reference string: \n");
    scanf("%d", &n);

    int pages[n];
    printf("Enter the reference string: \n");
    for(int i=0; i<n; i++){
        scanf("%d", &pages[i]);
    }

    printf("Enter number of frames: \n");
    scanf("%d", &frames);
    int frame[frames], time[frames];

    for(int i=0; i<frames; i++){
        frame[i] = -1;
        time[i] = 0;
    }

    int pageFaults = 0, pageHits = 0, counter = 0;

    for(int i=0; i<n; i++){
        int found=0;

        for(int j=0; j<frames; j++){
            if(frame[j] == pages[i]){
                found=1;
                pageHits++;
                counter++;
                time[j] = counter;
                break;
            }
        }

        if(!found){
            int LruIndex = 0;

            for(int j=1; j<frames; j++){
                if(time[j] < time[LruIndex]){
                    LruIndex = j;
                }
            }

            counter++;
            frame[LruIndex] = pages[i];
            time[LruIndex] = counter;
            pageFaults++;
        }
    }

    printf("\nTotal Page Faults: %d\n", pageFaults);
    printf("Total Page Hits: %d\n", pageHits);

    return 0;
}
```